WHZ Westsächsische Hochschule Zwickau
Hochschule für Mobilität

PORSCHE MOTORSPORT

Exposé: diploma thesis Porsche Motorsport EMA3

1. Title

Development of an software analysis tool for MATLAB/Simulink models and their generated C Code, providing a graphical signal dependency overview and computation data that can be used to optimize runtime and reduce cross-core data transitions on the Formula E VCU.

2. Motivation/Problem Statement

At Porsche Motorsport, a large part of the software deployed on the Formula E and LMDH vehicles is C code generated from MATLAB/Simulink models. These models are constantly updated and optimized by Porsche engineers to improve the vehicle's performance. If a certain function in one of these models needs to be changed, the engineer currently has to navigate through the model to track signal dependencies and understand the signal composition for all necessary inputs of the function to be improved. Tracing a signal through the model this way is time consuming and generates a cognitive overhead for these engineers. The engineers need a quick overview of individual signal composition and the origin of computation.

3. Objective

The objective of this thesis is to develop a software analysis tool for MATLAB/Simulink models that accepts an SLX model or the generated Code and Artifacts from MATLAB Embedded Coder as input. By using a user-selected signal (which can be an output of a block or a signal in the model), the tool will perform backward tracing of this signal through the model. The data gained from this analysis will be used to create a structural calculation tree that shows all blocks/values/signals/inputs that contribute to the signal. The result will be a readable map that instantly shows all relationships of the selected signal, containing additional information such as the origin of computation for each block. This gives engineers a fast way to understand, review, and document signal dependencies without manually navigating through the model.

4. Methodology/Procedure

- Analysis of generated C Code and Artifacts form Matlab Embedded Coder
- Tool development with the support of AI coding models
- The extent of AI generated code is at the student's discretion, but is encouraged by the employer.

5. Optional Extension / Benefits

Use of additional information gathered from the C code analysis to develop optimization strategies for multicore VCU code deployment (optional)
Recommendations for code placement across cores to reduce cross-core signal transfers.

6. Scope

- The focus is on the development of the analysis/mapping tool
- multicore optimization suggestions are supplementary/optional
- there is no automatic re-placement

7. Schedule

Major milestones:

- Literature review and familiarization with MATLAB Embedded Coder Artifacts
- Development of the analysis tool framework
- Implementation of backward tracing logic
- Visualization/map generation
- Tools ability to extract additional information used for multicore optimization
- Optional optimization strategy development

8. Support

Intern: Dr.-Ing. Stefan Lingl, Porsche EMA3
Extern: Prof. Dr.-Ing. Gernot Herbst, WHZ